

Nirja Dabhi

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Education

George Brown College

Advanced Diploma, Computer Programming and Analysis

Toronto, ON

Apr 2026

Experience

Tech Developer

Toronto, ON

HuskyHacks

Dec 2025 – May 2026

- Identified and documented a critical frontend–backend integration failure caused by inconsistent API error payloads, applying structured root-cause analysis to improve system reliability.
- Strengthened application reliability by implementing defensive error handling for malformed and empty server responses across shared services.
- Delivered peer-reviewed updates to a collaborative codebase using Git workflows (feature branches, PR reviews) within an agile team environment.

Peer Tutor

Toronto, ON

George Brown College

April 2024 – Dec 2025

- Supported students with debugging and resolving programming issues across Java, Python, and web development workflows.
- Explained core computer science concepts including control flow, OOP, data structures, and problem decomposition in a structured and accessible way.
- Developed strong communication skills by translating complex technical problems into actionable steps under time constraints.

Selected Projects

Gen AI Scholarship Aggregator | TypeScript, React, Chrome Extension, Supabase, Gemini AI, OpenRouter

- Built a retrieval-augmented matching system pairing Ontario students with scholarships, generating text-embedding-3-small vectors and serving semantic search over Supabase pgvector.
- Designed the ingestion pipeline end to end — Firecrawl scraping, Gemini-based structured extraction, schema validation, embedding and upsert — handling malformed and partial source records without failing the batch.
- Orchestrated eligibility filtering and match reasoning through Gemini, returning ranked results with explanations rather than raw similarity scores.
- Shipped as both a React web app and a Chrome extension.

Neural Network Experiments (CNN Regularization & Hyperparameter Studies) | Python, TensorFlow/Keras, NumPy, scikit-learn, pandas

- Diagnosed severe overfitting in a CNN on CIFAR-10 (50k images, 10 classes): the two-convolution model reached 100% training accuracy against 68% validation, a 32-point generalization gap, with validation peaking by epoch 5 of 50.
- Isolated undertraining as the cause of a 20% baseline through single-variable ablations across optimizer, depth, width, epochs, and batch size — increasing epochs 4→20 alone recovered accuracy to 83%.
- Designed 45°/135° directional line-detection kernels and applied edge operators through a NumPy convolution routine with zero-padding and pixel clamping, validating the feature extraction underlying Conv2D layers.

Pebble - Elderly Care Platform (Capstone) | JavaScript, Node.js, REST API, React Native, MongoDB, Expo

- Built RESTful backend services with role-based access control across three user roles (seniors, family members, caregivers), enforcing permission logic at the API layer for secure data isolation
- Collaborated in an agile team environment using Git feature branches and PR reviews, contributing to backend architecture and MongoDB schema design

Technical Skills

Languages: Python, SQL, TypeScript, JavaScript, Java, Bash

ML/AI: TensorFlow/Keras (CNNs, regularization, dropout), scikit-learn, pandas, NumPy, RAG architectures, embeddings and vector search, LLM integration (Gemini, GPT-4, Claude), prompt engineering

Data & Infra: PostgreSQL/pgvector, MongoDB, MySQL, Docker, Git/GitHub, CI/CD, Linux (WSL2)

Frameworks: React, React Native, Node.js, Express, Next.js